

Localizing and Beating the Out-of-Distribution Bottleneck in Vector Search: A Measured Collection of Routing, Coverage, and Systems Techniques

[TODO: authors]

Abstract

Scalable approximate nearest-neighbor (ANN) search routes a query to a few partition cells, scans quantized codes inside them, and re-ranks a shortlist. On *out-of-distribution* (OOD) workloads — where queries come from a different distribution than the database, as in cross-modal (text→image) retrieval under inner product — we first *localize the bottleneck*: a battery of scan-side and code-side improvements (wider SIMD, higher-resolution and rotated codes, RaBitQ, ADC routing) leave OOD recall/QPS essentially unmoved, while the achievable recall is capped by which cells get *routed to and covered*. We then present a collection of routing- and coverage-directed techniques, each measured in isolation: a one-parameter routing recalibration (γ) grounded in a cap-versus-ball selectivity argument; k -NN-graph coverage augmentation; a joint cross-level partition optimization; multi-assignment; and a corollary that *reverses* standard practice — once graph coverage is present, coarser cells win. Composed and evaluated under a same-hardware, tight-pairwise protocol against ScaNN at its official configuration, the engine attains higher QPS at matched recall@10 at 1M, 10M[[pending](#); , and 100M], and transfers to the streaming track. We report the negative results as first-class evidence for the localization, and treat the measurement methodology — which repeatedly caught mirages during development — as a contribution.

1 Introduction

[TODO: Motivate OOD vector search (RAG, cross-modal). The IVF+quant pipeline: route→scan→re-rank. Thesis sentence: on OOD the binding constraint is ROUTING/COVERAGE, not scan throughput or code fidelity – and we show this by measuring both sides. Then: a collection of coverage/routing techniques, each with its own measured effect, plus the informative negatives.]

Contributions (each measured in isolation).

- **A bottleneck localization** (§5): scan- and code-side levers do not move OOD; routing/coverage does. The negatives (§12) are the evidence.
- γ **routing recalibration** (§6), grounded in a cap-vs-ball selectivity argument (§4). *Effect: $\approx$ half the probes at matched OOD recall, zero query-time cost.*
- k -**NN-graph coverage** (§7). *Effect: composes with γ ; at 1M moves the ScaNN ratio $1.18 \rightarrow 0.98$.*
- **The coarse-cell corollary** (§8). *Effect: with graph coverage present, coarser cells win; at 10M $1.01 \rightarrow 0.85$.*
- **Joint cross-level partition optimization (tree-EM)** (§9). *Effect: $+0.6$ – 1.0 recall@10 points at 10M, QPS-neutral.*
- **Multi-assignment (SOAR)** (§10). *Effect: raises the routing recall ceiling to baseline parity.*
- **Systems levers for streaming** (§11): batch-parallel insert ($4.5\times$), low-live adaptive p (worst step $0.74 \rightarrow 0.94$), and a parallelization fix ($7.6\times$ at 8 threads).

- **A same-hardware evaluation methodology** (§13) and an honest scope of what the ratios do and do not claim.

2 Background: Standard Building Blocks

Our system is assembled from well-established components; we summarize them here so the contributions in §3–§10 can be stated as deltas against known practice.

Inverted file (IVF). Partition the database into C cells (the Voronoi regions of C centroids). At query time, *route* the query to its $p \ll C$ nearest centroids and scan only those cells. End-to-end recall is capped by whether the true neighbor’s cell is among the p probed — the routing question that this paper argues is the OOD bottleneck.

Hierarchical routing. Scoring all C centroids per query is $O(Cd)$, which dominates at scale (fine C is needed to keep cells small). A *tree* of centroids — coarse k -means over C_0 centroids, then fine k -means within each — reduces routing to $O(LC^{1/L})$ for an L -level tree (FAISS uses an HNSW graph over the centroids for the same purpose). We use hierarchical k -means.

Product quantization (PQ). Split each vector’s d dimensions into m subspaces and quantize each subspace to a small codebook (16 codewords for 4-bit PQ). Approximate distances are computed by lookup-and-add over per-subspace tables (ADC). With 4-bit codes the table lookup is a register shuffle (`vpshufb`), enabling the SIMD “fast-scan” kernels (Quick-ADC, ScaNN, FAISS FastScan). *Anisotropic* / score-aware quantization (ScaNN) weights the quantization loss to preserve inner-product ranking rather than reconstruction.

Exact re-ranking. The quantized scan is used only to select a shortlist; exact distances (int8 or float, read for the few shortlisted vectors) re-order it. The shortlist depth t trades recall for cost.

Multi-assignment. Replicating each point into its top- a_0 cells (a space/time tradeoff) raises the chance a query’s neighbor lands in a probed cell.

The OOD workload. The big-ANN OOD track (text2image) pairs *image*-embedding base vectors ($d=200$) with *text*-embedding queries under inner product, with a float ground truth. Base norms are concentrated but not constant (≈ 0.81 – 0.99); queries are unit-norm. This distribution gap between base and queries is what breaks L_2 -trained routing (§6).

3 System Overview

Figure 1 shows the pipeline; we describe the build and query paths, then delineate standard vs. contributed components (Table 1).

Build path.

1. **Quantize.** Float base $\rightarrow$ int8 via a single global scale ($127/\max|x|$); the int8 base is the native scan/route representation.
2. **Train the routing tree.** Hierarchical k -means: C_0 coarse centroids, then fine centroids within each coarse cell to K_f leaves total (2-level; generalizes to L levels).
3. **Joint refinement (tree-EM, §9).** Optionally re-optimize all levels jointly (E: reassign sample points to their nearest leaf via the beam descent; M: recompute every level’s centroids).

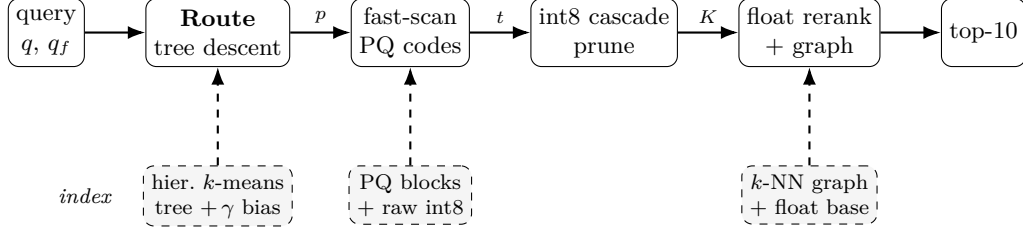


Figure 1: Query pipeline (top) and the offline-built index it consumes (bottom, dashed). Edge labels are the shrinking candidate set: p probed leaves $\rightarrow$ top- t pool $\rightarrow$ K cascade survivors $\rightarrow$ top-10. γ calibration, the k -NN graph union, and the coarse-cell tree geometry are this work’s contributions (Table 1).

4. **Multi-assign (SOAR, §10).** Assign each point to a_0 leaves with spill that covers the residual direction of the first.
5. **Encode.** 4-bit anisotropic PQ ($m=d/2$ subspaces); pack codes into cell-contiguous 16-row blocks laid out for the `vpshufb` fast-scan. Store the raw int8 vectors in cell order too (cache-warm re-rank gathers).
6. **Calibrate & augment.** Precompute the per-cell γ routing bias (§6); build the base-side k -NN graph sidecar (§7).

Query path.

1. **Route (tree descent).** Score the C_0 coarse centroids (int8, VNNI kernel), keep the top- b_0 ; expand to their fine children, score those with the γ bias added, and return the top- p leaves.
2. **Scan.** Fast-scan the 4-bit PQ codes of the p probed cells (`vpshufb` ADC with int16 accumulation for ~ 12 -bit selection resolution; optional AVX-512 64-wide), collecting a bounded top- t pool.
3. **Cascade re-rank.** Deduplicate the pool by original id (needed under multi-assignment); exact-rescore the t survivors in int8 (VNNI dot); prune to the top- K ($K=16$).
4. **Float re-rank + graph union.** Compute exact float inner products for the K survivors (reading the original float vectors) unioned with the k -NN graph neighbors of the top- M ; return the top-10.

Algorithms 1–3 state the pipeline precisely. All routing scores use the calibrated form $s_\gamma(q, c) = \gamma\|c\|^2 - 2q \cdot c$ (§6); the constant $\|q\|^2$ is dropped, and smaller s_γ means a better cell. Leaf centroids are $\{c_j\}_{j=1}^{K_f}$ with precomputed bias $b_j = (\gamma - 1)\|c_j\|^2$ so that $s_\gamma(q, c_j) = \|c_j\|^2 - 2q \cdot c_j + b_j$ is one dot product plus an additive constant.

4 Theoretical Grounding (Scoped)

[TODO: (i) Cap-vs-ball under concentration: additive ball filters are non-selective in high dimension (everything within $R + r$), angular/spherical-cap filters remain selective; k -means on centered data realizes a data-dependent angular partition. (ii) This frames γ : the routing score $\gamma\|c\|^2 - 2q \cdot c$ interpolates $\gamma=1$ (additive ball / L_2) to $\gamma \rightarrow 0$ (pure inner product / cap), so the theory predicts $\gamma < 1$ is needed precisely in the concentrated OOD regime – which we confirm. (iii) Two principles we build on: route-precise/rank-cheap, and multi-assignment as a space/time tradeoff. Explicitly state we do NOT claim the recursive-tree polylog result (refuted at $\leq 10M$ scale).]

5 Localizing the OOD Bottleneck

[TODO: Frame the diagnostic: pool-recall ceiling (re-rank the whole probed pool) vs scan-limited recall. Show scan is already extracting the pool (no scan headroom), so recall is set by routing/coverage. This motivates attacking routing (§5-9) and predicts scan/code levers won’t help (§12, the negatives).]

Algorithm 1: Build

Input: float base $X_f \in \mathbb{R}^{n \times d}$; tree sizes C_0, K_f ; multi-assign a_0 ; calibration γ ; EM rounds R ; graph degree k

Output: index $(\{Q_0\}, \{c_j\}, b, \text{PQ blocks, raw}, G)$

```

1  $\sigma \leftarrow 127 / \max_i \|X_f[i]\|_\infty$ ;  $X \leftarrow \text{int8}(\sigma X_f)$ ; // global-scale quantize
2  $\{Q_0\} \leftarrow k\text{-means}(X, C_0)$ ;  $\{c_j\} \leftarrow \text{fine } k\text{-means within each coarse cell to } K_f \text{ leaves}$ ;
3 for  $R$  rounds // tree-EM (§9); optional do
4   E: assign each sample to its nearest leaf via  $\text{Route}(\cdot, 1)$ ;
5   M: recompute  $\{Q_0\}, \{c_j\}$  from the assignments;
6 foreach point  $x_i$  // SOAR multi-assign (§10) do
7   assign  $x_i$  to its top- $a_0$  leaves; spill covers the residual of the 1st;
8 per leaf: encode members as 4-bit anisotropic PQ, pack into 16-row vpshufb blocks; store raw int8
   in leaf order;
9  $b_j \leftarrow (\gamma - 1) \|c_j\|^2$  for all  $j$ ; // routing bias
10  $G \leftarrow \text{base-side } k\text{-NN graph (self-search; §7)}$ ;
11 return index  $(\{Q_0\}, \{c_j\}, b, \text{PQ blocks, raw}, G)$ ;
```

Algorithm 2: Route (tree descent) – returns the p best leaves

```

1 Procedure  $\text{Route}(q, p)$ :
2    $S_0 \leftarrow \text{top-}b_0 \text{ coarse cells minimizing } s_\gamma(q, Q_0)$ ; // int8/VNNI kernel
3    $\mathcal{L} \leftarrow \text{fine children of } S_0$ ;
4   score each  $c_j \in \mathcal{L}$  by  $\|c_j\|^2 - 2q \cdot c_j + b_j$ ;
5   return the  $p$  leaves of  $\mathcal{L}$  with smallest score;
```

6 γ : One-Parameter Routing Recalibration

[TODO: Per-cell additive i32 bias $(\gamma-1)\|c\|^2$ precomputed at build; zero per-query cost. Fit once per dataset ($\gamma \approx 0.5$ on text2image, 1M and 10M). Ablate $\gamma \in \{0.4, 0.45, 0.5, 0.55, 0.6\}$.]

Effect. Reaches matched OOD recall at $\approx$ half the probes (p : 54 $\rightarrow$ 29 at 1M; $\approx 2.5\times$ probe cut at 10M). Zero query-time cost.

7 k -NN-Graph Coverage Augmentation

[TODO: Union the graph neighbors of the top- M shortlist members into the exact re-rank set (base-side IP-kNN sidecar; built by ScaNN self-search or engine self-query, $\sim 92\%$ edge agreement). Recovers deep neighbors a shallow probe missed. Composes additively with γ (partial overlap, compound ~ 0.97).]

Effect. At 1M, composed with γ , the ScaNN ratio drops 1.18 $\rightarrow$ 0.98 (the first sub-1 result); at 10M it lets t_{surv} halve at matched recall.

8 Routing Cost and the Coarse-Cell Corollary

8.1 A routing cost model

Consider an L -level tree over K_f leaves with cell counts $C_0 < C_1 < \dots < C_{L-1} < C_L = K_f$ and per-level beam widths $b_0, \dots, b_{L-2}$ (the number of cells expanded when descending from level i to level $i+1$). Routing a query costs, in centroid distance evaluations,

$$\text{route}(q) = C_0 + \sum_{i=1}^{L-1} b_{i-1} \frac{C_i}{C_{i-1}}, \quad (1)$$

Algorithm 3: Query

Input: int8 query q , float query q_f ; probes p ; pool depth t ; prune K ; graph fan-in M

```

1  $\mathcal{C} \leftarrow \text{Route}(q, p)$ ;
2  $\mathcal{P} \leftarrow \emptyset$ ; // bounded top- $t$  pool
3 foreach  $cell \in \mathcal{C}$  do
4   fast-scan its 4-bit PQ codes (vpshufb ADC, int16 accum)  $\rightarrow$  approx. scores;
5   keep the  $t$  best in  $\mathcal{P}$  (threshold-bounded collect);
6  $\mathcal{P} \leftarrow$  dedup  $\mathcal{P}$  by original id ; // needed under  $a_0 > 1$ 
7  $\mathcal{K} \leftarrow$  top- $K$  of  $\mathcal{P}$  by exact int8 IP (VNNI dot) ; // cascade prune
8  $\mathcal{R} \leftarrow \mathcal{K} \cup \bigcup_{r \in \text{top-}M(\mathcal{K})} G[r]$  ; //  $k$ -NN-graph coverage union
9 return top-10 of  $\mathcal{R}$  by exact float IP  $\langle q_f, x_r \rangle$ ;
```

Table 1: Components of the pipeline: standard building blocks vs. this work.

Component	Basis	Our delta
int8 IVF partition	standard	—
Hierarchical k -means tree	standard (FAISS-style)	fast int8/VNNI descent kernel
4-bit anisotropic PQ	ScaNN / FastScan family	int16-resolution + 32/64-wide fast-scan
Multi-assignment (SOAR)	ScaNN-derived	dedup-before-cap correctness fix
int8 $\rightarrow$ float re-rank cascade	folklore	the specific $K=16$ composition
γ routing calibration	ours	the OOD miscalibration fix
k -NN-graph coverage union	ours	composition with γ + float re-rank
Coarse-cell geometry knee	ours	reverses “finer is better”
Tree-EM joint optimization	ours	cross-level partition objective
Batch-insert / adaptive- p (streaming)	ours	§11

since the coarse level scores all C_0 centroids and level i expands the b_{i-1} best parents from level $i-1$, each contributing C_i/C_{i-1} children. The scan then reads $\text{scan}(q) = p \cdot (n/K_f) \cdot a_0$ candidates (p probed leaves $\times$ points/leaf $\times$ multi-assignment). For a 2-level tree (1) is $C_0 + b_0 K_f / C_0$, minimized at $C_0 = \sqrt{b_0 K_f}$ to give $2\sqrt{b_0 K_f} = O(\sqrt{K_f})$; with L levels and balanced branching this falls to $O(L K_f^{1/L})$. The two costs pull oppositely in K_f : finer leaves shrink the scan (n/K_f) but grow the routing. The beams b_i are the “routing aggression” knob — wider beams retain the true cell with higher probability at linear cost — and OOD queries, which route poorly, need wider beams (§14).

Table 2 gives geometry and routing cost by scale. Two levels suffice through 10M; at 100M a 2-level tree at its optimal $C_0 = \sqrt{b_0 K_f} \approx 16,400$ would cost $\approx 33,000$ evals, whereas the 3-level tree roughly halves this (14,336) — so the third level is introduced only at 100M, where routing is a large enough share of the query to matter.

8.2 The corollary

[TODO: Sweep cell granularity at 10M with the graph on: fine (Kf=262144), medium (65536), coarse (32768); medium/coarse WIN. Bracket the knee with four builds.] The cost model (1) explains *why*. Without graph coverage, recall demands fine K_f (small leaves make the true neighbor’s leaf likelier to be among the p probed), paying $O(\sqrt{K_f})$ routing. The k -NN-graph union (§7) supplies that coverage *independently* of K_f , so the tree can use *coarser* leaves — cheaper routing — at fixed recall. The optimum is a knee: too fine wastes routing, too coarse inflates the scan ($p \cdot n/K_f$). Empirically the knee sits near 150 points/leaf at 10M, and both finer and coarser builds lose to it.

Effect. At 10M, moving from the fine-cell to the 152-point-cell geometry takes the ScaNN ratio $1.01 \rightarrow 0.85$; routing evals/query drop $\sim 12,000 \rightarrow 6,144$.

Table 2: Tree geometry and per-query routing cost (Eq. 1) by scale. Beams are aggressive ($b_0 \approx 12\text{--}25\%$ of coarse cells) because OOD queries route poorly. [\[pending: 100M validated via a scaling-law sweep \(§8.1\).\]](#)

scale	L	K_f	cell counts	beams	route evals	pts/leaf
1M	2	16,384	$C_0=768$	$b_0=96$	$768 + 96 \cdot 21 = 2,816$	61
10M	2	65,536	$C_0=4096$	$b_0=128$	$4096 + 128 \cdot 16 = 6,144$	152
100M	3	524,288	$C_0=2048, C_1=32768$	$b_0=512, b_1=256$	$2048 + 512 \cdot 16 + 256 \cdot 16 = 14,336$	190

9 Joint Cross-Level Partition Optimization (Tree-EM)

[\[TODO: EM over the hierarchical partition: E-step reassigns to nearest leaf via the beam descent \(objective = query-time search\), M-step recomputes every level jointly. Cheap-beam E-step. Additive with \$\gamma/\text{graph}\$.\]](#)

Effect. +0.6–1.0 recall@10 points at 10M at matched probe, QPS-neutral (build-time only).

10 Multi-Assignment (SOAR)

[\[TODO: Replicate each point into \$a_0\$ cells; SOAR spill covers the residual direction of the first cell. A dedup fix \(rerank by orig id before the survivor cap\) was required to realize it. Raises the routing coverage ceiling.\]](#)

Effect. $a_0=2$ reaches OOD coverage parity with the baseline partition; higher a_0 extends the high-recall frontier at a storage/scan cost (quantified in table).

11 Systems Techniques (Streaming)

[\[TODO: msturing-30M-clustered final_runbook; recall@10 under 1h + 8GB. Three levers: \(i\) batch-parallel insert \(assignment parallelized over the batch, order-preserving, bit-identical\); \(ii\) low-live adaptive \$p\$ \(grow probes so expected candidates \$\geq\$ threshold when the live set is small\); \(iii\) a parallelization fix \(the batched re-rank driver ran chunks serially -> RAYON never engaged\). Plus a compliance re-architecture: first-insert-batch cold start + periodic live-set retraining.\]](#)

Effect. Batch insert: 30M inserts 1818s $\rightarrow$ 406s ($4.5\times$), recall-identical. Adaptive p : worst step 0.737 $\rightarrow$ 0.944. Parallelization fix: $7.6\times$ at 8 threads. Compliant recall@10 = [\[pending: final NQ=10000\]](#), peak <8GB.

12 Negative Results (Evidence for the Localization)

[\[TODO: Report each with its measured null/loss:\]](#)

- **Query-aware routing** (train cells on the query distribution): *worse* than base-trained (0.76 vs 0.89 @ matched p) — undertrained + the cells where queries are dense don’t distribute base points well.
- **ADC routing** (4-bit-scan the centroids): recall-neutral but *no speed win* — the exact VNNI router is already at the compute floor.
- **Anisotropic/OPQ-rotated coarse codes for rank preservation**: do *not* make coarse codes rank-preserving; reorder depth is set by code bit-rate, and rotation/ η move it < 0.01 . Coverage-via-coarsening collapses (a scann-coverage-matched dpb=50 config yields recall 0.09).
- **RaBitQ**: no ranking edge over PQ at matched bytes.
- **Wider SIMD** (AVX-512 vpermw / 64-wide): scan is memory-bound, not shuffle-bound; $\leq +7\%$, sometimes negative (downclock/cross-lane).
- **Finer cells at matched probe**: lose recall (half the probe fraction).

Table 3: OOD text2image, recall@10-matched QPS ratio (ScaNN/ours; < 1 = our engine faster). Same-hardware, tight-pairwise, opponent at official config. [\[pending: 100M row\]](#).

Scale	1-thread	8-thread	recall@10 gate
1M	0.755	0.756	0.905
10M	0.827	0.833	0.901
100M	[pending:]	[pending:]	[pending:]

Table 4: Ablation ladder at 10M (ScaNN/ours ratio; each row adds one lever).

Configuration	ratio
γ only	1.24
+ graph coverage	1.01
+ tree-EM	0.92
+ coarse-cell geometry (knee)	0.85
+ pristine confirmation (20 rounds)	0.83

- **PCA/LeanVec dim reduction on OOD:** near-isotropic embeddings; dead.

These cluster on the scan/code side and all fail; the §5–9 levers cluster on routing/coverage and all succeed — the localization made concrete.

13 Evaluation Methodology

We report *same-hardware ratios*, never cross-machine QPS. All comparisons run on one physical machine; each competitor is measured in the same session, seconds apart, at its own best operating point.

Protocol. For a target recall ($\text{recall@10} \geq 0.90$), we (i) fix each system’s operating point by an independent sweep, requiring both to clear the recall gate against the *same* exact float ground truth; (ii) alternate the two systems round to round (order flips) so neither owns a load window; (iii) take best-of- k within a round and report the *median of per-round ratios*; (iv) recompute recall from raw ids, never trusting a cached number. On a shared machine, load drift between two builds minutes apart exceeds the effects being measured, so only interleaved same-window ratios and load-independent recall are trusted.

Warm-resident semantics; pristine confirmation. Both systems measured warm and resident (leaderboard serving); an early cold-spawn asymmetry is corrected by scoring each engine’s warm block. Headline numbers pause all other first-party jobs (SIGSTOP) and confirm across ≥ 20 rounds.

The baseline at its best. ScaNN uses its *official* big-ANN recipe (e.g. 40k leaves at 10M, anisotropic AH/LUT16, residual quant, bfloat16 re-order) with its own leaves/re-order sweep including corner configs. A win depending on a handicapped baseline is quarantined; we describe one (an under-leaved cached index) and its correction.

What we do not claim. Same-hardware ratios on one (shared) machine, not a standardized-VM submission; ScaNN is the *measured* opponent — gaps to binary-only leaderboard systems are *inference* from the public leaderboard.

14 Results

[TODO: Add per-lever isolation deltas; 8-thread scaling; streaming table (recall, peak mem, wall vs 1h budget). Note the ablation ordering is the actual development arc.]

15 Related Work

[TODO: ScaNN (anisotropic VQ), DiskANN/HNSW (graphs), FAISS IVF-PQ, big-ANN challenge, RaBitQ. Position OOD-specific routing as the gap; distinguish γ from residual quantization (orthogonal) and from query-aware training (shown to fail).]

16 Limitations

[TODO: Single (shared) machine; γ fit per dataset; the coarse-cell corollary is demonstrated on IP/OOD data and may not transfer to in-distribution L_2 ; the gap to binary-only leaderboard systems is inference; filter/sparse tracks untouched; standardized-VM submission is future work and the gating experiment for a leaderboard claim.]